

Algebraic Topology: a beginner's course

This lecture:

**AlgTop9: Applications of
Euler's formula and graphs**

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Lecture 9: Applications of Euler's formula and graphs



Lecture 9: Applications of Euler's formula and graphs

Thm There are at most 5 Platonic solids.

Proof. Assume have a regular polytope
 $n = \# \text{ edges on each face (regular)}$
 $m = \# \text{ edges that meet at a vertex}$

Formula

onic

polytope

(regular)

a vertex

Count edges: $E = \frac{F \times n}{2}$

Count vertices $V =$



$$E = \frac{F \times n}{2}$$

so $F\left(\frac{n}{m} - \frac{n}{2} + 1\right) = 2$

so also $\boxed{6 > n}$

possibilities
are

$(n, m) = (3, 3)^T$

$(3, 4)^T$

$(3, 5)$

$(4, 3)$

$(5, 3)$

$V = \frac{F \times n}{m}$

$$F = \frac{4m}{2n - mn + 2m}$$

must have $2n - mn + 2m > 0$

or $\boxed{2(n+m) > nm}$

$$2n > m(n-2) \quad m, n \geq 3$$

$$\frac{2n}{n-2} > m \geq 3 \Rightarrow 2n > 3n - 6$$

$\boxed{6 > n}$

Count edges: $E = \frac{F \times n}{2}$

so $F\left(\frac{n}{m} - \frac{n}{2} + 1\right) = 2$

Count vertices $V = \frac{F \times n}{m}$

$$F = \frac{4m}{2n - mn + 2m}$$

$$V - E + F = 2$$

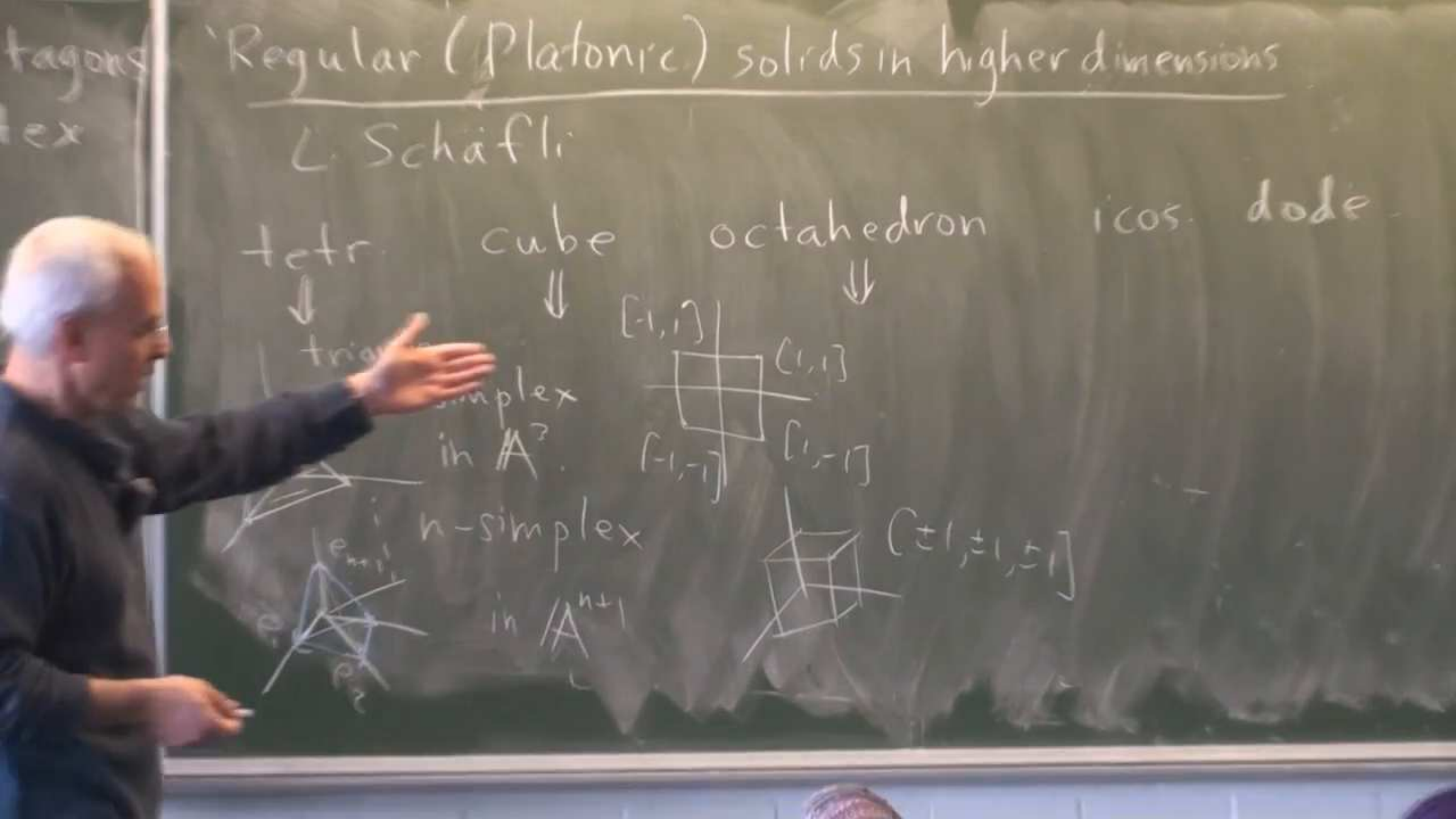
must have $2n - mn + 2m$

$$\frac{F \times n}{m} - \frac{F \times n}{2} + F = 2$$

or $\boxed{2(n+m) > nm}$

$$2n > m(n-2) \quad m, n \geq 3$$

$$\frac{2n}{n-2} > m \geq 3 \Rightarrow 2n$$



Regular (Platonic) solids in higher dimensions

L. Schäfli

tetr.



triangle

cube

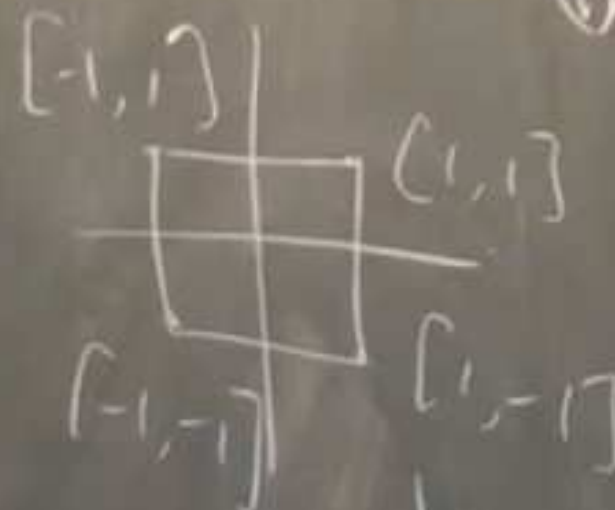


octahedron

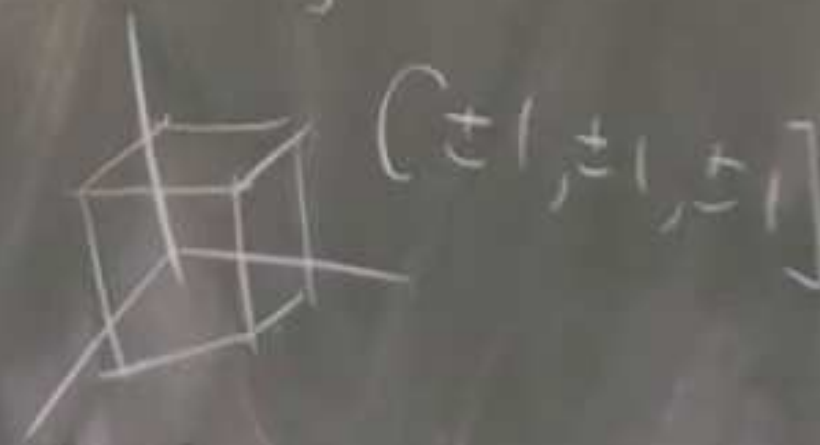


icos. dode

simplex
in A^3



n -simplex
in A^{n+1}

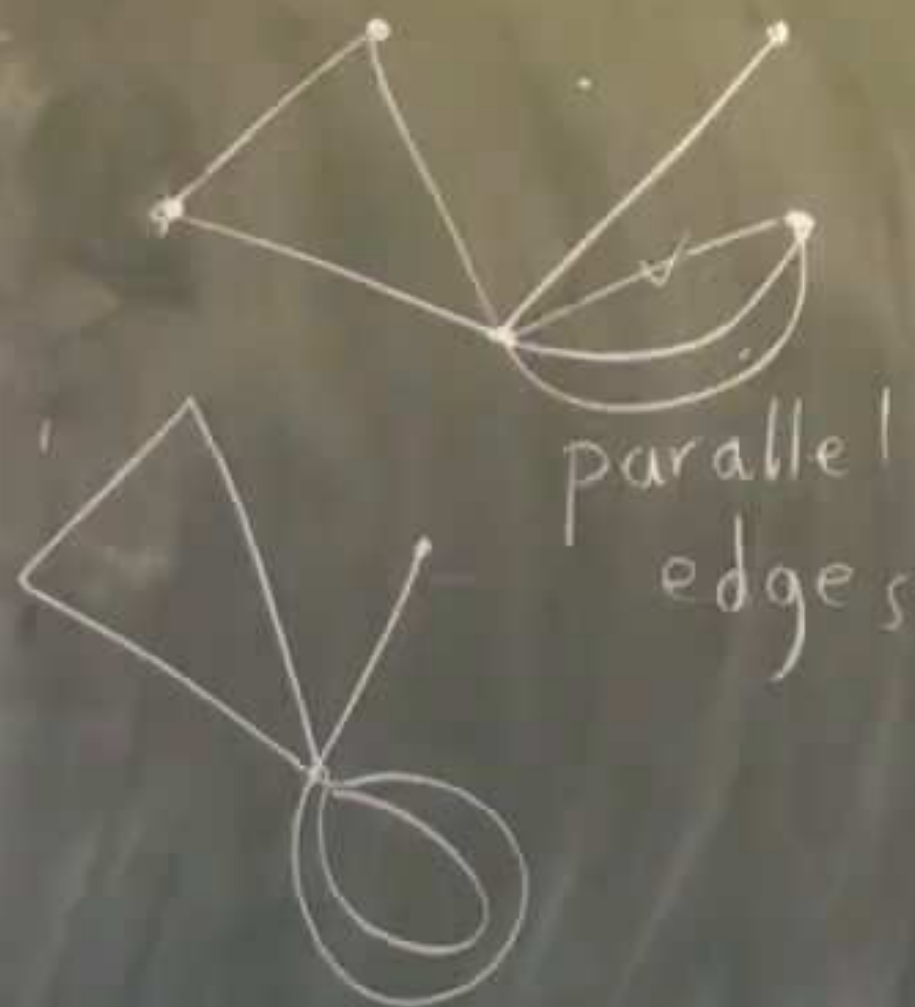


Graphs

ell

120

Graphs



simple graph . no loops
no parallel edges

assume connected

$$V = 5$$

$$E = 8$$

$$F = 5 \text{ (outside included)}$$

$$V - E + F = 2$$

Thm For

$$V - E + F$$

Proof -

If it is a

Then $E \rightarrow$

If it is joined
shrink it so V

$$E \rightarrow E - 1$$